Particle swarm optimization algorithm for solving airline crew scheduling problem

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***Abstract*—In air transport, the cost related to crew members presents one of the most important cost supported by airline companies. The objective of the crew scheduling problem is to determine a minimum-cost set of pairings so that every flight leg is assigned a qualified crew and every pairing satisfies the set of applicable work rules. In this paper, we propose a solution for the crew scheduling problem with Particle Swarm Optimization (PSO) algorithm, this solution approach is compared with the Genetic Algorithm (GA) for both crew pairing and crew assignment problems which are the two part of crew scheduling problem.**

***Keywords* – Aircraft, Crew pairing, Crew assignment, Optimization, Particle Swarm Optimization**

I. INTRODUCTION

Air transport has experienced competition. This situation obliges airlines to improve existing approaches and develop new planning techniques. Airline companies seek to optimize the planning of schedules of crew in order to control the operating costs, the crew costs are one of the biggest components of the operating costs. The problem is to cover the cost of flights with planned over a given time horizon, by crews formed cockpit crew (pilot, copilot) and cabin crew (hostess, stewards). Each crew from the base to which he is assigned, connects a number of flights and returns to base. This sequence of flights back to the base is called rotation. The construction of rotations of an airline company is constrained by the international and national labor regulations, and the limited availability of resources. These constraints make it difficult to solve the problem, the use of models and optimization software for this problem allows companies to make financial gains. The crew scheduling problem is decomposed into two subproblems. First, the crew pairing problem is to construct, for duty period, a series of rotations from all flight segments. Then, the crew assignment problem consists in assigning rotations by adding vacation days and other activities to construct for each period about one month. The crew pairing and the crew assignment must also consider the rules defined by the aviation safety. Planning for a company of passenger air transport is a complex activity and its decomposition into sub-problems yielded good solutions quickly. We want to search a minimum cost set of feasible pairings, in this set every flight is covered exactly once.

Crew pairing and crew assignment problem have been the subject of several studies over the last 40 years. There are several review articles that define and present the approaches to solve this problems as Desaulniers et al (1998) [13], Barnhart et al. (1999) [6], Gopalakrishnan and Johnson (2005) [18] and Klabjan (2005) [25]. Moreover, Saddoune et al. (2010) [28] have developed a method to solve the problem of construction simultaneously of rotations and the problem of building monthly blocks for pilots. Furthermore, Jarrah and Diamond (1997) [21] introduced a heuristic approach that uses partitioning column generation a priori where the objective function is to minimize the total cost and unassigned rotations (open time). Also, Campbell et al. (1997) [8] have developed a heuristic for FedEx two phases whose objective is to minimize the number of blocks and the number of rotations and unassigned to maximize the purity of these blocks.

Christou et al. (1999) [9] used a genetic algorithm in two phases to build anonymous blocks at Delta Air Lines The first phase of the algorithm seeks to produce blocks while the second phase completes the blocks by adding uncovered rotations. Weir and Johnson (2004) [34] proposed a three-phase approach where each represents a given optimization problem. The objective of this approach is to build monthly blocks providing quality of life for employees and ensuring the greatest possible regularity in the arrangement of tasks having the effect of reducing fatigue. Boubaker et al. (2010) [7] proposed a model for constructing approximate partitioning monthly blocks for pilots.

Glanert (1984) [17] proposed a method of assigning a rotation of maximum priority to an employee of maximum priority. Ryan (1992) [30] proposed an approach that consists in generating a priori blocks using heuristics. Gamache and Soumis (1998) [16] developed a method of column generation that solves the linear relaxation of the optimality problem. Day and Ryan (1997) proposed a method that gradually improves the solution. ElMoudani et al. (2001) [14] suggested a heuristic approach to solve a problem by considering two criteria: operating costs and overall satisfaction" crew members. Dawid et al. (2001) [12] introduced an implicit enumeration heuristic implemented in the SWIFTROSTER algorithm.

Gamache et al. (1998) [16] proposed a sequential approach based on a list ordered by seniority of employees. This approach has the advantage of building a schedule for each employee, considered optimal given previously assigned to the oldest hourly employees. An improvement of this approach has been discussed in Gamache et al. (2007) [15] using a model and a graph coloring algorithm, tabu search to determine whether there exists at least one of feasible solution. Achour et al. (2007) [1] presented for the first time an exact method to solve the personalized problem with seniority. The approach solves a sequence of integer programs by column generation, each program being associated with an employee. Cordeau et al. (2001) [10] presented a resolution approach based on Benders decomposition to solve routing problems and building aircraft rotations considering instances of medium size. Klabjan et al. (2002) [26] proposed a model that incorporates partially the problems of flight planning, aircraft routing and construction crew rotations. Zeghal and Minoux (2006) [35] proposed an approach for building blocks for pilots, officers and instructors. Crainic and Rousseau (1987) [11] proposed a heuristic approach to solve the Crew pairing problem. Gershkoff (1989) on the other hand used a heuristic to generate subsets of flight services. Anbil et al. (1992) developed a method to select a subset of rotations in order to obtain a better estimate of the global optimum of the problem of crew pairing. Also, Graves et al. (1993) proposed a system (generation and optimization) to construct a minimum cost rotations. Hoffman and Padberg (1993) [20] proposed a heuristic algorithm that Despite the fact that there are few publications in the literature which developed algorithms to solve the crew scheduling problem based on meta-heuristics. This approach useful in some cases as we can see from the following papers.

Firstly, Ozdemir and Mohan (2001) [29] presented this problem as a flight graph model and applied genetic algorithm to solve it. Also, Souai and Teghem (2009) [32] proposed an approach based on genetic algorithm to integrate airline crew pairing and rostering problem: Their purpose is to solve these two problems simultaneously. Limlawan et al. (2011) [27] proposed a particle swarm optimization to solve the airline crew rostering problem by assigning appropriate balance workload for each crew member. Furthermore, Zeren and Ozkol (2012) [5] proposed a solution to the crew pairing problem based on genetic algorithms with construction of a genetic operator. The paper of Shen et al. (2013) [31] presented a genetic algorithm based on an adaptive evolutionary approach for the crew scheduling problem in public transport. Moreover, Azadeh et al. [4] (2012) presented a particle swarm optimization algorithm for solving the crew scheduling problem. Azadeh et al. (2013) [3] also proposed an PSO algorithm combined with a local search heuristic for solving the crew scheduling problem. Jian and Chou [22] proposed a a multi-objective genetic algorithm to solve the train crew pairing problem in railway companies.

Recently, some papers tended to hybridize meta-heuristics with other approaches. For instance, the paper of Aydemir-Karadag et al. (2013) [2] aimed to propose a hybrid method which uses meta-heuristic to reduce the search space and then uses an exact method to have the solution. Also, Hanafi and Kozan (2014) [19] used a constructive heuristic coupled with the Simulated Annealing algorithm to solve the problem of railway crew scheduling.

The use of meta-heuristic in this context is justified by the fact that crew scheduling is an NP-hard problem.

The organization of this paper is as follows. In Section 2, the problem of crew scheduling is described. The section 3 presents the experiments. The conclusion of the paper is presented in section 4.

II. PROBLEM STATEMENT

The crew scheduling problem was often decomposed in two parts: The crew pairing and crew assignment (rostering).

A. *Classical formulation of the problem*

The crew pairing problem can mostly be formulated as a set covering problem (SCP) or set partitioning problem (SPP). The SPP formulation can be expressed as follows:

$$\min \sum_{j \in P} c_j x_j \tag{1}$$

$$s.t \begin{cases} \sum_{j \in P} a_{ij} x_j = 1 \ \forall i \in F \\ x_j \in \{0,1\} \qquad \forall j \in P \end{cases} \tag{2}$$

F : Set of all flight legs
P : Set of all legal pairings
c_j : cost of pairing j
a_{ij} : 1, if flight leg i is covered by pairing j. 0, otherwise.
x_j : is equal to 1 if pairing p is included in the solution, and 0 otherwise.

As mentioned before, the problem can also be modeled as SCP, while modifying Constraints. The first constraint will be in this case as shown below:

$$\sum_{j \in P} a_{ij} x_j \geq 1 \ \forall i \in F \tag{3}$$

This constraint allows deadheading that is crew can travel as passengers and reposition to the crew home base or duty base to maintain his duty.

The second stage of the crew scheduling process is the crew assignment (or crew rostering). The basic model for the crew assignment was given by Gamache and Soumis (1998) [16].

$$\min \sum_{k \in K} \sum_{s \in S^k} c_s^k x_s^k \tag{4}$$

$$s.t \begin{cases} \sum_{k \in K} \sum_{s \in S^k} \gamma_p^s x_s^k \geq n_p \ \forall p \in P \\ \sum_{s \in S^k} x_s^k = 1 \qquad \forall k \in K \\ x_s^k \in \{0,1\} \end{cases} \tag{5}$$

K : the set of crew members of a given type.
S^k : the set of work schedules that are feasible for employee $k \in K$.
P : the set of dated pairings to be covered.
n_p : represents the minimum number of crew members that must be assigned to pairing $p \in P$.
γ_p^s : is 1 if pairing $p \in P$ is included in schedule s and 0 otherwise.
x_s^k : equals 1 if schedule $s \in S^k$ is assigned to employee $k \in K$ and 0 otherwise.
c_s^k : the cost of schedule $s \in S^k$ for employee $k \in K$ represents the schedule cost.

The main purpose for the optimization of crew scheduling is to minimize the total cost. This cost can be calculated by the summing the different crew pairing costs.

B. Resolution of the problem by the Binary PSO algorithm

The particle swarm optimization (PSO) metaheuristic is an evolutionary approach proposed by Kennedy and Eberhart in 1995 [23]. The swarm consists of individual agents named particles, and represented by vectors, where each particle corresponds to a potential solution of an optimization problem.

In PSO, Each particle i has two vectors: the velocity vector and the position vector: The particles are updated according to itself previous best position and the entire swarm previous best position. That is, particle i adjusts its velocity ν_i and position x_i in each generation according to the following formula:

$$\begin{cases}\nu^{n+1} = \omega\nu^n + c_1 r(p^n - x^n) + c_2.r.(p_g^n - x^n) \\ x^{n+1} = x^n + \beta.\nu^n\end{cases} \quad (6)$$

where ν^n and x^n are the current velocity and position of the particle. p^n represents the best previous position of particle i. p_g^n represents the best position among all particles in the population. r_1 and r_2 are two independently uniformly distributed random variables with a range. The efficiency of the PSO depends on its ability to perform global search of the search space and local search as well.

The discrete (binary) version of the PSO algorithm was defined by Kennedy and Eberhart in 1997 [24]. The update of particles are done in the same manner that in the continuous version of PSO. But, the binary PSO used the sigmoid function which can be defined as follows:

$$sig(x) = \frac{1}{1 + exp(-x)} \quad (7)$$

Therefore, the decision variable can be defined as in equation (8):

$$x^{n+1} = \begin{cases}0\, if\; r \leq sig(\nu)^{n+1} \\ 1\, otherwise\end{cases} \quad (8)$$

with r is a uniform random number in the range of [0,1].

C. The proposed approach

In this paper, we follow the approach of decomposing the problem into two parts. The first one is the pairing generation, where a number of legal pairings are generated and the cost of each pairing is calculated. The second one is the optimization of the solution, where a subset of the generated pairings is selected, so that every ight leg in the schedule is included in at least one pairing and the total cost is minimized.

As we have seen from the literature review, PSO has been applied in two cases to the crew scheduling problem. The first for the crew rostering (2012) [4] and the second for the crew pairing (2013) [3]. But, the crew scheduling is frequently modeled with decision variables which are binary variables. Thereby, it will be important to apply the binary version of PSO to this kind of problem.

At the best of our knowledge, the binary PSO algorithm hasn't been yet applied for solving the mathematical model of the crew scheduling problem. Therefore, the PSO algorithm will be applied in this paper and compared to GA.

III. Experiments

To perform optimization of the Crew Scheduling problem with Metaheuristics, we used the Matlab software. The Genetic Algorithm is predefined in the optimization toolbox of Matlab. To build the PSO algorithm for Crew Scheduling, we developed a binary version of the PSO algorithm.

in order to evaluate the efficiency of the PSO and GA algorithms, we have tested them on eight generated hypothetical problems that consist of which includes a number of flights and a number of pairing that has been selected as test data. Two optimization algorithms are tested. The first is the GA algorithm and the binary version of PSO which is proposed in this paper.

A. Parameters setting

The two algorithms have been executed on the same platform: Intel Core i5 PC, 1.79 GHz with 1 GB of RAM under Windows 7 operating system (Virtual machine). Algorithm has been run over 7 instances for the crew pairing problem. For each instance, we performed 8 run using PSO and GA and we noted the max and min values.

For the two algorithms we consider a population size of 50 (particles for PSO and chromosomes for GA).

Fig. 2. Presentation of the fitness function

The following table presents the minimum of the fitness function (best cost) and the CPU time (expressed in seconds and rounded to the unit).

Instance (I)	Flight leg	Number of pairing
1	5	5
2	10	5
3	5	9
4	10	9
5	20	10
6	20	20
7	40	20

Table 1. Parameters setting

B. Computational results

For each of the two optimizations algorithms (GA an PSO), we display the results obtained for the crew pairing and assignment problems.

I	PSO			GA		
	Min	Max	CPU	Min	Max	CPU
1	103	103	0.13	103	103	3.5
2	98	146	0.14	98	146	1.9
3	149	149	0.11	99	151	3.6
4	143	156	0.15	98	147	3.5
5	194	298	0.22	94	103	3.8
6	195	300	0.23	147	150	3.6
7	403	647	0.37	143	149	3.5

Table 2. Numerical results by problem size

	PSO		GA	
	Average cost	CPU	Average cost	CPU
1	103	1	103	2
2	146	0.13	103	1.9
3	149	0.11	151	3.6
4	149	0.15	101	3.5
5	245	0.22	98	3.7
6	248	0.23	147	3.6
7	596	0.37	147	3.5

Table 3. Tendency of results

C. Discussion

We can see from the results of the previous section that the GA and PSO give approximately the same cost (fitness function). In other words, they may minimize the total cost in the same level. Also, the PSO algorithm may give a small lower CPU time as presented in the table. Thus, the utilization of the PSO is justified for the crew scheduling problem.

Moreover, the PSO algorithm proposed in this study can be improved by better defining parameters (particles...). In this case, the PSO may give better results than GA.

IV. CONCLUSION

In this paper, we presented solution methodology for the airline crew scheduling problem. This problem can be decomposed into two sub-problems: The crew pairing and crew assignment. To optimize these sub-problems, a solution is proposed by the metaheuristic algorithm Particle Swarm Optimization and compared to Genetic Algorithm. computational results have shown the effectiveness of the PSO in solving this problem.

Future research should attempt to improve the PSO algorithm to obtain better results for the resolution of airline crew scheduling problem and to add more constraints to this problem.

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